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Date: ___/___/2023

MTH603

Current Paper

Today: 15-08-2023

Q#1:- In Second Order Runge-Kutta method K_1 is given by

a) $K_1 = 2hf(x_n, y_n)$ b) $K_1 = hf(x_n, y_n)$

c) $K_1 = 3hf(x_n, y_n)$ d) None of these

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c) $K_1 = 3hf(x_n, y_n)$ d) None of these

Q#2:- The relationship between forward difference operator and the shift operator is given by

a) $\Delta = E-1$ b) $\Delta = E^{-1}-1$

c) $\Delta = E+1$ d) None of these

c) $\Delta = E+1$ d) None of these

Q3:- The magnitude of truncation error in Milne's Corrector formula is

a) $\frac{1}{90} h \Delta^4 y_0$ ✓ b) $\frac{1}{90} h \Delta^3 y_0$

c) $\frac{28}{90} h \Delta^3 y_0$ d) $\frac{28}{90} h \Delta^4 y_0$

Q#4:- To apply Simpson's $1/3$ rule, the number of interval must be

a) $\frac{1}{90} h \Delta^4 y_0$ b) $\frac{1}{90} h \Delta^3 y_0$

c) $\frac{28}{90} h \Delta^3 y_0$ d) $\frac{28}{90} h \Delta^4 y_0$

Q#4.- To apply Simpson's $1/3$ rule, the number of interval must be

a) 1 b) 2

c) 3 d) 5

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5- Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial Condition $y = 1.02$ at $t_0 = 0.02$. Using Euler's method, y at $t = 0.04$, $h = 0.02$ is

y

a) 3.0392

b) 1.0392

d) 0.0392

6- Adam-Moultan P-C method is a multistep method where we assume that the Solution to the given initial value problem is known in past ______ equally spaced points

a) 1 b) 2 c) 3 d) 4

a) 1 b) 2 c) 3 d) 4

7.

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

Value of $f'(0.5)$ using the two point equation.

a) -0.5 b) 0.5

d) -0.75

8. $\frac{dy}{dx} = 1-y, \; y(0)=0$ is an example of

a) An Ordinary differential b) A polynomial equation
c) A partial differential d) None of these

9. The global error of Simpson's 1/3 is same as the global error.

9- The global error of Simpson's $1/3$ is same as the global error.

a) Rectangular rule b) Trapezoidal

✓c) Simpson $\frac{3}{8}$ rule d) Runge-Kutta rule

10- When we apply Simpson's $3/8$ rule,

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a) Rectangular rule   b) Trapezoidal

✓ c) Simpson $\frac{3}{8}$ rule   d) Runge-Kutta rule

10- When we apply Simpson's 3/8 rule, the number of Interval is must be

a) 5   b) 7

✓ c) 9   d) 11

a) 5 b) 7

c) 9 d) 11

11. When we apply Simpson's 1/3 rule, the number of Interval is must be

a) Even b) Odd

c) d) None of thes

12. In 3rd ... order Runge-Kutta method

In forth Order Runge-Kutta method, K_3 is given by

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a) $K_3 = hf\left(x_n + \frac{h}{3}, y_n + \frac{K_2}{3}\right)$ b) $K_3 = hf\left(x_n - \frac{h}{2}, y_n - \frac{K_2}{2}\right)$

✓ c) $K_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right)$ d) $K_3 = hf\left(x_n - \frac{h}{3}, y_n - \frac{K_2}{3}\right)$

13- The Order of global error in

c) $K_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right)$ d) $K_3 = hf\left(x_n - \frac{h}{3}, y_n - \frac{K_2}{3}\right)$

13. The Order of global error in Simpson's $1/3$ rule is ____ than to the order of global error in Simpson's $3/8$ rule.

a) Equal b) Less

14. In Solving the differential equation
$y' = x+2y$, $y(1) = 2$
$h=1$, By Euler method $y(2)$ is calculated as $y_1 = y_0 + hf(\ )$

a) 1 b) 3 = 9 =

c) 5 d) 7

calculated as.

16. In Integral $\int_0^{\frac{\pi}{2}} \sin x\,dx$, by dividing the Interval into four equal parts, width of the interval

a) π b) $\frac{\pi}{2}$

c) $\frac{\pi}{4}$ ✓d) $\frac{\pi}{8}$

17. In Integral $\int^{2}$ ...

c) $\frac{\pi}{4}$ ✓ d) $\frac{\pi}{8}$

$\frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}$

17- In Integrating $\int_0^2 e^{2x} dx$, by dividing the interval into 8 equal parts width of the interval at

$\frac{2-0}{8}$

a) 0.125 ✓ b) 0.250

c) 0.500 d) 0.625

c) 0.500 d) 0.625

18- In Interval $\int_0^1 e^{2x}\,dx$, by dividing the Interval into ~~four~~ eight equal parts, width of the Interval is

a) ✓ 0.125 b) 0.250

c) 0.500 d) 0.625

c) 0.500 d) 0.625

19. In Second Order Runge-Kutta method K_2 is given by

a) $K_2 = hf\left[x_n + \frac{1}{3}h, y_n + \frac{1}{3}K_1\right]$ b) $K_2 = hf\left\{x_n - \frac{1}{3}h, y_n - \frac{1}{3}K_1\right\}$

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✓ c) $K_2 = hf\left\{x_n + \frac{1}{2}h, y_n + \frac{1}{2}K_1\right]$ d) $K_2 = hf\left[x_n - \frac{1}{2}h, y_n - \frac{1}{2}K_1\right]$

20- Newton's Forward Difference Formula is used for Interpolating the value of y

a) Near the end of the set value

b) A Short distance from y_0

c) Short distance backward y_0

d) Near the beginning of set

d) Near the beginning of set

21- The number of polynomial than can go through two fixed data points (x_1, y_1) and (x_2, y_2) is

a) 2 b) 1

c) Infinite d) 0

22- Next

22. Next Approximation to the root of the given equation can be found in

a) $f(2)=3$, $f(3)=4$ b) $f(2)=-3$, $f(3)=4$

c) $f(2)=-3$, $f(3)=-4$ d) None of these

23. Next Approximation to the root

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d) None of these

24. For a given table of values, the process of estimating the value of y, for any intermidiate value of x, is known as

a) Interpolation ✓ b) Extrapolation
c) Forward Difference d) Backward Difference

a) Interpolation b) Extrapolation
c) Forward Difference d) Backward Difference

25 The trunction error in Adam's predictor formula is _____ times more than that in corrector formula.

a) 10 b) 11
c) 12 d) 13

26- _____ lies in the category of Iterative method

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a) Bisection Method b) Regula-Falsi method

c) Muller Method d) All of the given

c) Muller Method   d) All of the given

27- If we retain n+1 term in Newton's forward Difference formula, we obtain a polynomial of degree ___

a) n+1   b) n+2   c) n   d) n-1

a) $n+1$ b) $n+2$ c) n d) $n-1$

28 p in Newton's forward difference formula is defind.

a) $ph = x - x_0$ b) $p = (x - x_0)h$

c) $P = (x + x_0)h$ d) None of these

None of These

29- Give the following data by Newton Divided Difference, the value $f(2,4)$ is

x	0	1	2	4
$f(x)$	1	1	2	5

√ a) 1.5 b) 3 c) 2 d) 1

30. Given the following data

x	1	2	-4
$f(x)$	5	-5	-4

which formula is useful in finding the Interpolationg polynomial.

a) Newton's backward Difference b) Lagrange's Interpolation

c) Newton's Forward Difference d) None of these

31. In Richardson's extrapolation method, the extrapolation process is repeated and accuracy is achieved, this is called extrapolation to the ___

a) Limit ✓   a) function

c) Arbitrary value of 'h'   d) None

c) Arbitrary value of 'h'   d) None

32- A forth Order differential equation can be reduced to a system of four 1 order ordinary differential equation.

a) First   b) Second

c) Third   d) Fourth

3?. The minimum Interval in which

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the root of the equation $x^2-3x+1=0$ lie is

a) (0,1) ✓ b) (1,2)

c) (2,4) d) (0,2)

33. If there are $(n+1)$ values of y Corresponding to $(n+1)$ values of x, then we can represent the function $f(x)$ by a polynomial of degree

a) $n+2$ b) $n+1$

c) n d) $n-1$

34- From the following table of values

x	1.3	1.5	1.7	1.9	2.1	2.3
y	2.9648	2.659	2.33	1.992	1.644	1.296

Estimation of $y'(1.3)$ will be done using the ~~forward~~ formula of differential

a) Backward Difference b) Central Difference

c) Forward Difference d) None

35-

a) Backward Difference b) Central Difference
c) Forward Difference d) None

35- In forth-order Runge-kutta method, K_4 is given by

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a) $K_4 = hf(x_n + 2h, y_n + 2k_3)$ b) $K_4 = hf(x_n - h, y_n - k_3)$
c) $K_4 = hf(x_n + h, y_n + k_3)$ d) None

36. Milne - predictor method is a multistep method where we assume that the solution to the given initial value problem is known in past ____ equally spaced point

a) 1 b) 2 c) 3 d) 4

37. In double Integration, both the Integration

37- In double Integration, both the Integration at different values of the Independent variable, we divide the interval ____ of size h, such that interval ____ Sub-interval of size k.

a) Equal, unequal b) Unequal, unequal

c) equal, equal d) Unequal, equal

Q#1:- Write Newton's backward Difference formula for Interpolation.

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$$y_x = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots \frac{p(p+1)(p+2)\cdots(p+n-1)}{n!}\nabla^n y_n + \text{Error}$$

Question #2:- $f(3)$ from the following data by Lagrange's

x	0	1	4
y	5	9	17

Solution:-

Lagrange Formula:-

Solution:-

Lagrange formula:-

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

$$f(3) = \frac{(3-1)(3-4)}{(0-1)(0-4)}(5) + \frac{(3-0)(3-4)}{(1-0)(1-4)}(9) + \frac{(3-0)(3-1)}{\quad}(11)$$

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 +$$

$$\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

$$f(3) = \frac{(3-1)(3-4)}{(0-1)(0-4)}(5) + \frac{(3-0)(3-4)}{(1-0)(1-4)}(9) + \frac{(3-0)(3-1)}{(4-0)(4-1)}(17)$$

$$= \frac{-2}{4}(5) + \frac{(-3)}{(-3)}(9) + \frac{(6)}{12}(17)$$

$$f(3) = \frac{(3-1)(3-4)}{(0-1)(0-4)}(5) + \frac{(3-0)(3-4)}{(1-0)(1-4)}(9) + \frac{(3-0)(3-1)}{(4-0)(4-1)}(17)$$

$$= \frac{-2}{4}(5) + \frac{(-3)}{(-3)}(9) + \frac{(6)}{12}(17)$$

$$= -2.5 + 9 + 8.5$$

$f(3) = 15$ Solution

Question #3:- Evaluate the Integral

$\int^{4} x^2 dx$ by using Trapezoidal

Question #3:- Evaluate the Integral $\int_0^4 x^2 dx$ by using Trapezoidal rule. Take $h=1$

Formula of Trapezoidal rule

$$\int_0^4 x^2 dx = \frac{h}{2}\left[y_0+y_4+2(y_1+y_2+y_3)\right]$$

x_0 = lower limit = 0 , $y_0 = x_0^2 = 0$

Formula of Trapezoidal rule

$$\int_0^4 x^2\,dx = \frac{h}{2}\left[y_0+y_4+2(y_1+y_2+y_3)\right]$$

x_0 = lower limit = 0 , $y_0 = x_0^2 = 0$

$x_1 = x_0+h = 0+1 = 1$, $y_1 = x_1^2 = 1$

$x_2 = x_1+h = 2$, $y_2 = x_2^2 = (2)^2 = 4$

$x_3 = x_2+h = 3$, $y_3 = x_3^2 = 9$

$x_4 = x_3+h = 4$, $y_4 = x_4^2 = 16$

$x_3 = x_2 + h = 3$, $y_3 = x_3^2 = 9$

$x_4 = x_3 + h = 4$, $y_4 = x_4^2 = 16$

$$\int_0^4 x^2\,dx = \frac{1}{2}\left[0 + 16 + 2(1+4+9)\right]$$

$$= \frac{1}{2}(44)$$

$= 22$ Required Solution

Question #4:- $y' = x^2 + y^2$, $y(0) = 1$, Using

$= \frac{1}{2}(44)$

$= 22$ Required Solution

Question #4:- $y' = x^2 + y^2$, $y(0) = 1$, Using Euler's method to find y at $x = 2$, $h = 2$

$y' = x^2 + y^2$, $y_0 = 1$, $x_0 = 0$

Euler's Formula

$y_1 = y_0 + h f(x_0, y_0)$

$$y_1 = 1 + 2\left[x^2 + y^2\right]$$

$$y_1 = 1 + 2\left[(0)^2 + (1)^2\right]$$

$$= 1 + 2(1)$$

$$y_1 = 3 \quad \text{Required Solution.}$$

Question #5:-

Find $y''(8)$ from the following backward difference table.

$y_1 = 3$ Required Solution.

Question #5 :-

Find $y'(8)$ from the following backward difference table.

x	y	∇y	∇y^2	$\nabla^3 y$	$\nabla^4 y$
1	1				
		7			
2	8		12		
		19			
3	27			6	0
			18		

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	1				
2	8	7	12		
3	27	19	18	6	0
4	64	37	24	6	0
5	125	61	30	6	0
6	216	91	36	6	0
7	343	127	42	6	0
8	512	169			

Solution

7	343	127	42	6	0
8	512	169			

Solution:-

Using Backward difference formula for $y''(x)$

$$y_n'' = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n\right]$$

$$y''(8) = \frac{1}{1}\left[42 + 6 + \frac{11}{12}(0)\right]$$

$$y''(8) = 48$$

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Question #6:- If $y' = t^2 + 2t + y$, then find

Question #6:- If $y' = t^2 + 2t + y$, then find the next two derivates in term of t and/or y.

Solution:-

$$y' = t^2 + 2t + y$$

Take derivate with respect to 'y'

$$y'' = 0 + 0 + y'$$

Solution:-

$$y' = t^2 + 2t + y$$

Take derivate with respect to 'y'

$$y'' = 0 + 0 + y'$$

Again, Derivate w.r.t "y"

$$y''' = y''$$

Now Take derivate with respect to "t"

$$y' = t^2 + 2t + y$$

Take derivate with respect to 'y'

$y'' = 0+0+y'$

Againe, Derivate w.r.t "y"

$y''' = y''$

Now Take derivate with respect to "t"

$y' = t^2+2t+y$

$y' = 2t+2$

$y''' = y''$

Now Take derivate with respect to "t"

$y' = t^2+2t+y$

$y' = 2t+2$

Again:

$y' = 2$

Question#7:- Find Newton's Divided Difference table for the following

Question#7:- Find Newton's Divided Difference table for the following

x	0	1	2	4	5
y	1	14	15	5	6

Solution:-

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1st Divided 2nd Diff 3rd Diff 4th Di

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x	$f(x)$	1st Divided	2nd Diff	3rd Diff	4th Diff
0	1				
		$\frac{14-1}{1-0}=13$			
1	14		$\frac{1-13}{2-0}=-6$		
		$\frac{15-14}{2-1}=1$		$\frac{-2+6}{4-0}=1$	
2	15		$\frac{-5-1}{4-1}=-2$		$\frac{1-1}{5-0}=0$
		$\frac{5-15}{4-2}=-5$		$\frac{2+2}{5-1}=1$	
4	5		$\frac{1+5}{5-2}=2$		
		$\frac{6-5}{5-4}=1$			
5	6				

Que. Evaluate the Integral $\int^{4}(x^2+x+2)dx$

4	5	$\frac{\ }{4-2} = \dots$ $\longrightarrow$	$\frac{1+5}{5-2} = 2$	$5-1$
5	6 $\rightarrow$	$\frac{6-5}{5-4} = 1$		

Q#8:- Evaluate the Integral $\int_0^4 (x^2+x+2)\,dx$ Using Simpson $\frac{1}{3}$ rule. Take $h=1$

Formul of Simpso $1/3$ rule

$$\int_0^4 (x^2+x+2)\,dx = \frac{h}{3}\left[y_0 + y_4 + 4(y_1+y_3) + 2(y_2)\right]$$

Q#8:- Evaluate the Integral $\int_0 (x^2+x+2)dx$
Using Simpson $\frac{1}{3}$ rule. Take $h=1$

Formul of Simpso 1/3 rule

$$\int_0^4 (x^2+x+2)\,dx = \frac{h}{3}\left[y_0+y_4+4(y_1+y_3)+2(y_2)\right]$$

x_0 = lower limit $= 0$, $y_0 = x_0^2+x_0+2 = 2$

$$\int_0^4 (x^2+x+2)\,dx = \frac{h}{3}\left[y_0+y_4+4(y_1+y_3)+2(y_2)\right]$$

x_0 = lower limit = 0 , $y_0 = x_0^2+x_0+2 = 2$

$x_1 = 1$, $y_1 = 4$

$x_2 = 2$, $y_2 = 8$

$x_3 = 3$, $y_3 = 14$

$x_4 = 4$, $y_4 = 22$

$$\int_0^4 (x^2+x+2)\,dx = \frac{1}{3}\left[2+22+4(4+14)+2(8)\right]$$

$$= \frac{1}{3}\left[24+72+16\right]$$

$$= 37.3$$ Required Solution.

Q#9:- Evaluate the Integral $\int_0^3 (x^2+2)\,dx$ Using Simpson $\frac{3}{8}$ rule. Take $h=1$

Solution:-

$$\int_0^3 (x^2+2)\,dx$$

Q#9:- Evaluate the Integral $\int^{3}(x^2+2)dx$ Using Simpson $\frac{3}{8}$ rule. Take $h=1.0$

Solution:-

$$\int_0^3 (x^2+2)dx$$

First of all

$x_0 =$ lower limit $= 0 \rightarrow y_0 = x_0^2+2 = 2$

$x_2 = 2 \qquad , \quad y_2 =$

$x_3 = 3 \qquad , \quad y_3 = 11$

$$\int_0^3 (x^2+2)\,dx = \frac{3h}{8}\left[y_0 + y_3 + 3(y_1+y_2)\right]$$

$$= \frac{3(1)}{8}\left[2 + 11 + 3(3+6)\right]$$

$$= \frac{3}{8}\left[40\right)$$

15 Required Solution.